MILITARY SPECIFICATION

SQUIB, ONE AMPERE, NO FIRE

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers one type of one ampere, no fire squib.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-406	Plastics, Organic, General Specifications, Test Methods
QQ-W-343	Wire, Electrical and Non-Electrical, Copper and Uninsulated
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-676	Boxes, Set-Up, Paperboard

Military

MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-202	Test Methods for Electronic Component Parts
MIL-STD-350	Jolt Test for Use in Production of Fuzes
MIL-STD-352	40 Foot Drop Test for Use in Production of Fuzes
MIL-STD-353	Transportation Vibration Test for Production of Fuzes

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MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MS 20003-2	Indicator, Humidity Card, Three Spot, Impregnated Areas

(When requesting any of the above documents, give the title and complete designation of the item shown above. Copies of this specification and other unclassified specifications, standards, and publications required by contractors in connection with specific procurement functions may be obtained from the Commanding Officer, Navy Supply Depot (CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

PUBLICATIONS

Applied Mathematics Panel

AMP Report Number 101.1R SRG-P Number 40	Statistical Analysis For a New Procedure in Sensitivity Experiments

(Applications for copies should be addressed to Applied Mathematics Panel, National Defense Research Committee, Bruceton Laboratories, Bruceton, Pennsylvania.)

Interstate Commerce Commission

49 CFR 71-90	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402. Orders for the above publication should cite the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the squib shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining

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that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. After satisfactorily passing all the preproduction tests specified herein, no changes in raw materials for production shall be made without prior written approval of the procuring activity.

3.2 Data requirements. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

3.3 Materials

3.3.1 Energetic materials. The squib shall be a relatively insensitive electrical initiator designed to give uniform and rapid functioning times over extreme temperature ranges. The squib shall be initiated by a lead thiocyanate based composition and shall contain approximately 100 milligrams of metal oxidant composition as its booster and case charges.

3.3.2 Leadwires. The basic squib may be constructed with a variety of leadwire lengths and types. The actual length and type of leadwire shall be specified in the purchase order. The type of leadwires permissible include, but are not limited to, the following:

a. Solid, 22 gage, tin coated, copper conductor in accordance with QQ-W-343, polyamide insulated with a dimension over the insulation not less than 0.038 nor more than 0.041 inch.
b. Type E-24 stranded leadwire in accordance with MIL-W-16878/4 except that the conductor shall consist of not less than 19 strands and the diameter over the insulation shall be not less than 0.038 nor more than 0.041 inch.
c. Type E-26 stranded leadwire in accordance with MIL-W-16878/4 except that the conductors shall consist of not less than 19 strands.
d. Stranded leadwire identical to 3.3.2 b except that the insulation shall be of the polyamide type rather than the fluorocarbon specified in MIL-W-16878/4, Type E.

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3.3.3 Dimensions. The dimensions of the squib shall be as shown on Figure 1.

3.4 Functioning characteristics

3.4.1 Resistance. The squib shall have a continuous circuit with a resistance of 0.17 ± 0.03 ohm (exclusive of leadwire resistance).

3.4.2 No-fire current. The squib shall not fire when a current of 1.0 ampere is applied to the squib circuit for one minute.

3.4.3 All-fire current. The squib shall fire when a current of 2.0 amperes minimum is applied to the squib circuit.

3.4.4 Functioning time current. At 70 degrees F the squib shall have a functioning time of less than 6.0 milliseconds when fired at 5.0 amperes.

3.4.5 Electrostatic sensitivity. There shall be no detectable degradation of performance of the squib, as specified herein, after subjection to a static discharge across the bridgewire from a 500 micromicrofarad capacitor when tested in accordance with 4.6.4.

3.4.6 Insulation resistance. The insulation resistance of the loaded squib shall be not less than 5 megohms when subjected to a direct-current potential of no less than 250 volts for no less than 2 seconds.

3.4.7 Autoignition temperature. The squib shall not fire after exposure to a temperature of 240 ± 10 degrees F for no less than 8 hours.

3.4.8 Drop. The squib shall not fire and shall be safe to handle and dispose of after subjection to the 40-foot drop test specified in MIL-STD-352.

3.4.9 Jolt and vibration. The squib shall function normally without any detectable degradation in performance after exposure to the following conditions:

(a) Jolt testing per MIL-STD-350

(b) Transportation vibration per MIL-STD-353

3.5 Component requirements. During the manufacture of the squib, the following in-process tests shall be performed on the squib components:

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3.5.1 Acetone extraction. The phenolic squib plugs shall withstand the acetone extraction test with a loss of not more than 5 percent in weight.

3.5.2 Bridgewire. After being soldered to the plug terminal posts, the bridgewire shall withstand a force of 50 grams applied normal to the axis at the center of the wire for no less than 10 seconds.

3.5.3 Squib compositions. The three squib compositions employed in the manufacture of the squib shall have the following nominal weights and minimum calorific outputs:

Composition	Weight (milligrams)	Minimum Output (calories/gram)
Bead composition	5	650
Booster composition	55	550
Main case charge	45	1050

3.5.4 Leadwires. Each leadwire in the plug assembly shall withstand a pulling force of 10 pounds without showing evidence of loosening upon visual inspection.

3.6 Workmanship. The squib shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the squib shall be classified as follows:

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a. In-Process inspection (see 4.5)
b. Preproduction inspection (see 4.6)
c. Quality conformance inspection (see 4.7)

4.3 Sampling

4.3.1 Sampling plans. Unless otherwise specified, sampling plans and procedures used in the testing program shall be in accordance with MIL-STD-105 or MIL-STD-414 as specified for each test in Table I.

4.3.2 Inspection lot. A lot of squibs shall be defined as a group of squibs manufactured in a single manufacturing sequence under homogeneous conditions and presented for inspection at one time. All components utilized in the manufacturing process, including all chemical compositions, shall have been from homogeneous lots.

4.4 Test conditions. All tests performed under this specification shall be performed at the following standard ambient conditions unless otherwise stated in the test procedure:

a. Temperature	77 degrees F ± 20 degrees F
b. Relative humidity	12 to 90 percent
c. Barometric pressure	30 ± 2 inches of mercury

4.5 In-Process inspections

4.5.1 Acetone extraction test. Sufficient samples of representative plug assemblies shall be selected from each lot to conduct the acetone extraction test in accordance with Method 7021 of L-P-406. All of the plugs represented by the tested samples shall be considered to be defective if the extraction value exceeds 5 percent by weight and production shall be discontinued until such changes are made to enable the plug subassemblies to meet the requirements of 3.5.1.

4.5.2 Leadwire tightness test. The sample of squib plugs selected from each lot shall be held in a test fixture that exerts a pull of 10 pounds on each leadwire. Upon visual inspection, any plugs in which the leadwire shows evidence of loosening shall be considered to be defective.

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4.5.3 Bridgewire strength test. All bridged plug subassemblies in the production lot shall be mounted in a fixture that exerts a minimum load of 50 grams normal to the axis of the wire at its center. Individual assemblies that fail to pass this test shall be rejected.

4.5.4 Squib composition test. Two individual 2-gram samples from each new batch of bead composition, booster composition, or main case charge shall be dispensed for analysis of calorific output after mixing. Samples shall be burned in a Parr Oxygen Bomb (or equivalent) in accordance with the manufacturer's instructions. If the average of the two samples does not exceed the minimum values specified in 3.5.3 the entire batch shall be rejected and excluded from further processing.

4.6 Preproduction inspection. A sample of 220 squibs shall be taken for preproduction inspection. Ten samples shall be subjected to the auto-ignition test (4.6.1) and ten samples shall be subjected to the drop test (4.6.2). The remaining 200 squibs shall be subjected to the non-destructive tests (4.7.1), and then subjected to a static discharge test (4.6.4). All squibs which fire from the electrostatic discharge shall be considered defective. Following the static discharge test, all 200 squibs shall be subjected to jolt and transportation vibration (4.7.2.2.) as specified in MIL-STD-350 and MIL-STD-353, respectively. Following the jolt and transportation vibration tests, 100 squibs shall be subjected to the Bruceton all fire, no fire analysis (4.6.3). Then the remaining 100 squibs shall be fired as specified in 4.7.2.1.

4.6.1 Autoignition. A sample of 10 squibs selected from the lot shall be subjected to a temperature of 240 ± 10 degrees F for no less than 8 hours. Any squib that fires shall be considered to be defective.

4.6.2 Drop test. A sample of 10 squibs selected from the lot shall be subjected to a 40-foot drop test in accordance with MIL-STD-352. For acceptance, all squibs shall not fire and shall be safe to handle and dispose of.

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4.6.3 Bruceton all fire no-fire test. One hundred squibs shall be subjected to the Bruceton all fire no-fire analysis (AMP report number 101.1R SRG-P number 40). Fifty squibs shall be subjected to the Bruceton analysis at -35 degrees F and fifty squibs shall be subjected to the Bruceton analysis at +165 degrees F. Amperage increments shall be 0.05 and the dwell time shall be 5 seconds for all the Bruceton tests. In the Bruceton analysis, the test shall be acceptable if the Bruceton results yields a probability of 0.001 or less at a 60 percent confidence level that a squib would ignite under the specified no-fire conditions of 1.00 ampere or fail to ignite under the specified 2.0 ampere firing conditions, continuously applied for 5 seconds.

4.6.4 Static discharge test. A sample of squibs selected from the production lot per Table I shall first be subjected to a static discharge from a 500 micromicrofarad capacitor charged to 6500 ± 350 volts, and then fired as in 4.7.2.1. The functioning time shall be recorded. All squibs which fire from the electrostatic discharge or which fail to fire within the required functioning time shall be considered to be defective.

4.7 Quality conformance inspection

4.7.1 Non-destructive. Squibs which pass any of the tests in this group may be used in samples for further testing but shall not be returned to the lot. Any squibs which fire during this test shall be considered to be defective.

4.7.1.1 No-fire. A sample of squibs selected from the lot per Table I shall be tested at a current of 1.00 ± 0.05 ampere for not less than 15 seconds. The current shall be supplied from a 24 to 28 volt direct-current source using suitable series limiting resistors. Acceptance shall be based upon an AQL of 0.40 percent.

4.7.1.2 Circuit continuity and resistance. This test shall be performed on 100 percent of production. Each squib shall be tested at a current not exceeding 25 milliamperes applied to the ends of the leadwires. For acceptance, the squib shall have a continuous circuit with a resistance of

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not less than 0.14 ohm or greater than 0.20 ohm (exclusive of leadwire resistance). Squibs not passing this test shall be considered to be defective and eliminated from further processing on an individual basis.

4.7.1.3 Insulation resistance. The insulation resistance of the squib shall be demonstrated on a sample of squibs from the production lot selected per Table I, and tested in accordance with Method 302 of MIL-STD-202. A direct-current potential of 250 volts shall be applied between the shunted leadwires and the case for a minimum of 2.0 seconds. Squibs having an insulation resistance of less than 5 megohms shall be considered to be defective. If the sample fails to meet this requirement, the lot shall then be subjected to a 100 percent inspection with all defective squibs being eliminated on an individual basis from the lot.

4.7.2 Destructive. The following destructive tests shall be performed in accordance with the sampling plans of Table I.

4.7.2.1 Functioning time current test. A sample of squibs selected from the production lot per Table I shall be fired at ambient temperature at a test current of 5.0 ± 0.1 amperes. The functioning time for all squibs shall be recorded. Acceptance shall be based upon the sample satisfying an AQL of 1.0 percent, single specification limit, MIL-STD-414, Table B-1, with an upper limit for functioning time of 6.0 milliseconds at 5.0 ± 0.1 amperes.

4.7.2.2 Jolt and transportation vibration tests. A sample of squibs selected from the production lot per Table I shall be subjected to the jolt and transportation vibration conditions specified in MIL-STD-350 and MIL-STD-353 respectively, and then fired as in 4.7.2.1. The functioning time of all squibs shall be recorded. All squibs which fire during jolt and vibration testing, or which fail to fire within the required functioning time during post-test firing shall be considered to be defective.

4.7.3 Visual examination. Visually examine each container in a lot for conformance with 3.6 and Section 5.

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4.8 Acceptance criteria

4.8.1 In-Process

4.8.1.1 In-Process components. Lots of components shall be accepted with the sampling plans in Table I. Rejected lots can be screened to remove individual defectives and then resubmitted.

4.8.1.2 Preproduction. Failure of any sample to meet any preproduction requirements shall be cause for rejection of the total preproduction sample.

4.8.3 Quality conformance. Failure of any sample to meet any requirement of this specification shall be cause for rejection of the sample. Acceptance of the lot shall be in accordance with Table I.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified by the procuring activity, the preservation, packaging, and packing shall be in accordance with 49 CFR 71-90.

5.1.1 Squibs with leadwires of less than 2 feet. The squibs shall be assembled in groups of 25 or 50 and secured with tape or insulated wire. A total of 100 squibs shall be packaged in a set-up or folding paperboard box which meets as a minimum the board requirements for weight of contents as contained in Specification PPP-B-676 or Specification PPP-B-566.

5.1.2 Squibs with leadwires of more than 2 feet. The leads shall be coiled into a circle with a maximum diameter of 6 inches and each squib and coil inserted into a paper tube approximately 3 inches in length and 1/2 inch in diameter. A total of 50 to 100 squibs shall be packaged in a set-up or folding paperboard box as specified in 5.1.1.

5.1.3 Moisture protection. For interplant shipment or extended storage, each paperboard box shall be sealed in a moisture proof wrapper in accordance with MIL-P-116, method II (b). Three spot humidity indicator cards in accordance with MS 20003-2 shall be enclosed within each barrier external to the paperboard container. If all three spots turn pink on any one

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humidity card, the squibs shall not be used.

6. NOTES

6.1 Intended use. The squibs covered in this specification are intended for use in igniters used to ignite solid propellant rocket motors.

6.2 Service life. The squib shall be capable of meeting all of the requirements of Section 3 at any time within a period of 3 years from the date of manufacture.

6.3 Ordering data. Procurement documents shall specify, but not be limited to, the following information:

a. Title, number, and revision letter of this specification
b. Place of inspection, if source inspected
c. Lot size
d. Place of inspection
e. Disposition of test data
f. Type and length of leadwires

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TABLE I. - Summary of Squib Tests

Test	Requirement	Sampling			Test Method
		MIL-STD	Level	AQL	
In-Process tests	3.5	--	--	--	4.5
Acetone extraction	3.5.1	*	*	*	4.5.1
Bridgewire strength	3.5.2	*	*	*	4.5.3
Squib mix	3.5.3	*	*	*	4.5.4
Leadwire tightness	3.5.4	105	II	0.065	4.5.2
Preproduction tests					4.6
Autoignition	3.4.7	*	*	*	4.6.1
Drop	3.4.8	*	*	*	4.6.2
Bruceton analysis	3.4.2, 3.4.3	*	*	*	4.6.3
Resistance and circuit continuity	3.4.1	*	*	*	4.7.1.2
No-Fire	3.4.2	*	*	*	4.7.1.1
Insulation resistance	3.4.6	*	*	*	4.7.1.3
Electrostatic sensitivity	3.4.5	*	*	*	4.6.4
Jolt	3.4.9	*	*	*	4.7.2.2
Transportation	3.4.9	*	*	*	4.7.2.2
Functioning time	3.4.4	**	**	**	4.7.2.1
Quality conformance					4.7
Resistance and circuit continuity	3.4.1	*	*	*	4.7.1.2
No-Fire	3.4.2	105	II	0.40	4.7.1.1
Functioning time	3.4.4	414	IV	1.0	4.7.2.1
Jolt	3.4.9	105	II	1.0	4.7.2.2
Transportation	3.4.9	105	II	1.0	4.7.2.2
Insulation resistance	3.4.6	105	II	1.0	4.7.1.3

*Sampling plan as stated in test method.
**The remaining 100 units shall be evaluated for conformance with MIL-STD-414, (AQL, 1.0 percent), Table B-1 for an upper limit of 6.0 milliseconds (see 4.6).

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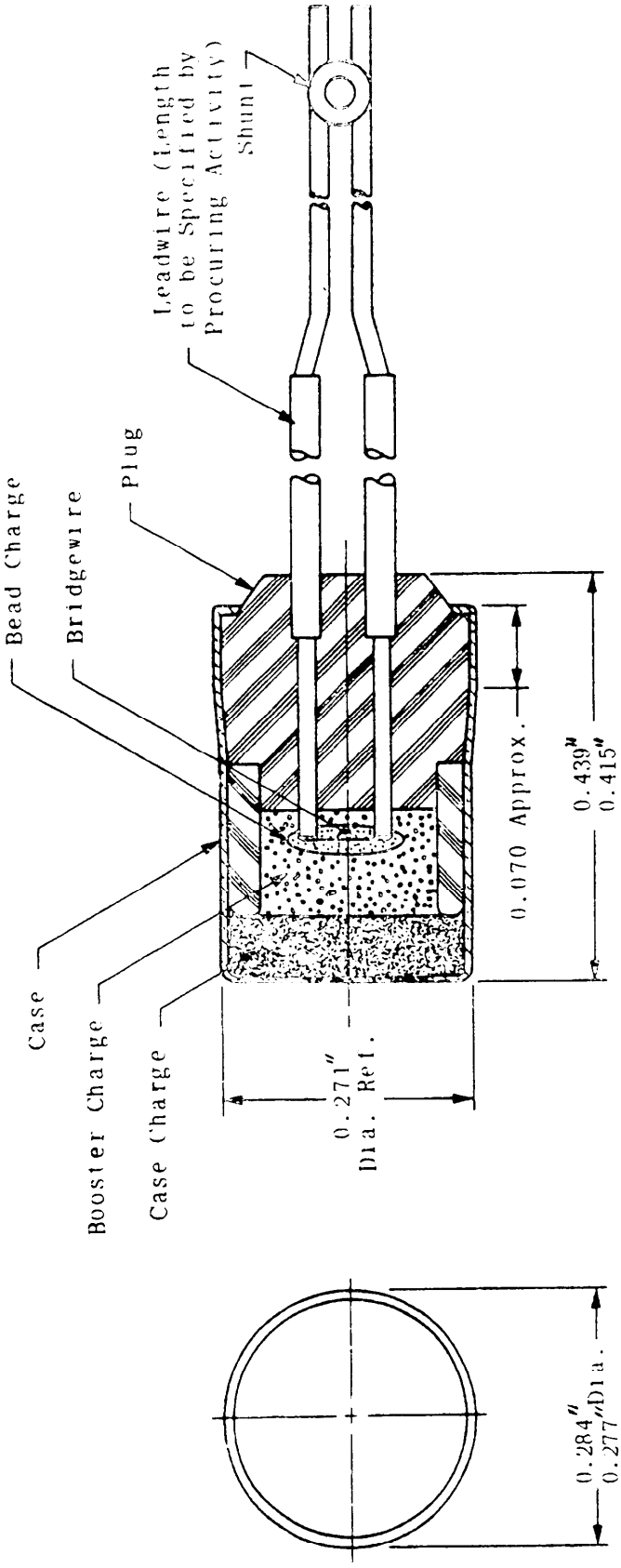


Figure 1. Squib One Ampere, No Fire

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

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